

Titanic Dataset Exploratory Data Analysis (EDA)

Objective: Extract insights from the Titanic dataset using statistical and visual exploration techniques. **Tools:** Python (Pandas, Matplotlib, Seaborn) **Dataset:** Titanic passenger data from Seaborn library

```
In [14]: !pip install pandas seaborn matplotlib
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load Titanic dataset from seaborn
file_path=r"C:\Users\LIBERARY\Downloads\train.csv"
titanic = pd.read_csv(file_path)

# Display first few rows
titanic.head()
```

```
Requirement already satisfied: pandas in c:\users\library\appdata\local\programs\python\python313\lib\site-packages (2.3.2)
Requirement already satisfied: seaborn in c:\users\library\appdata\local\programs\python\python313\lib\site-packages (0.13.2)
Requirement already satisfied: matplotlib in c:\users\library\appdata\local\programs\python\python313\lib\site-packages (3.10.6)
Requirement already satisfied: numpy>=1.26.0 in c:\users\library\appdata\local\programs\python\python313\lib\site-packages (from pandas) (2.3.3)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\library\appdata\roaming\python\python313\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\library\appdata\local\programs\python\python313\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\library\appdata\local\programs\python\python313\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\library\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cyclor>=0.10 in c:\users\library\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\library\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (4.60.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\library\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\users\library\appdata\roaming\python\python313\site-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\library\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\library\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (3.2.5)
Requirement already satisfied: six>=1.5 in c:\users\library\appdata\roaming\python\python313\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
[notice] A new release of pip is available: 25.1.1 -> 25.2
[notice] To update, run: python.exe -m pip install --upgrade pip
```

Out[14]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500



Data Overview

Let's examine the dataset's structure, types, and missing data.

```
In [15]: # Data info
         titanic.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId 891 non-null   int64
1   Survived    891 non-null   int64
2   Pclass      891 non-null   int64
3   Name        891 non-null   object
4   Sex         891 non-null   object
5   Age         714 non-null   float64
6   SibSp       891 non-null   int64
7   Parch       891 non-null   int64
8   Ticket      891 non-null   object
9   Fare        891 non-null   float64
10  Cabin       204 non-null   object
11  Embarked    889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

Identify which columns have missing data.

```
In [16]: # Summary statistics
         titanic.describe()
```

Out[16]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204200
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
In [17]: titanic
```

Out[17]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.25
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0	PC 17599	71.28
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.92
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.10
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.05
...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.73

891 rows × 12 columns



```
In [20]: # Value counts for categorical columns
print("Passenger Class Counts:\n", titanic['Pclass'].value_counts())
print("\nSex Counts:\n", titanic['Sex'].value_counts())
print("\nEmbarked Counts:\n", titanic['Embarked'].value_counts())
```

Passenger Class Counts:

Pclass

3 491

1 216

2 184

Name: count, dtype: int64

Sex Counts:

Sex

male 577

female 314

Name: count, dtype: int64

Embarked Counts:

Embarked

S 644

C 168

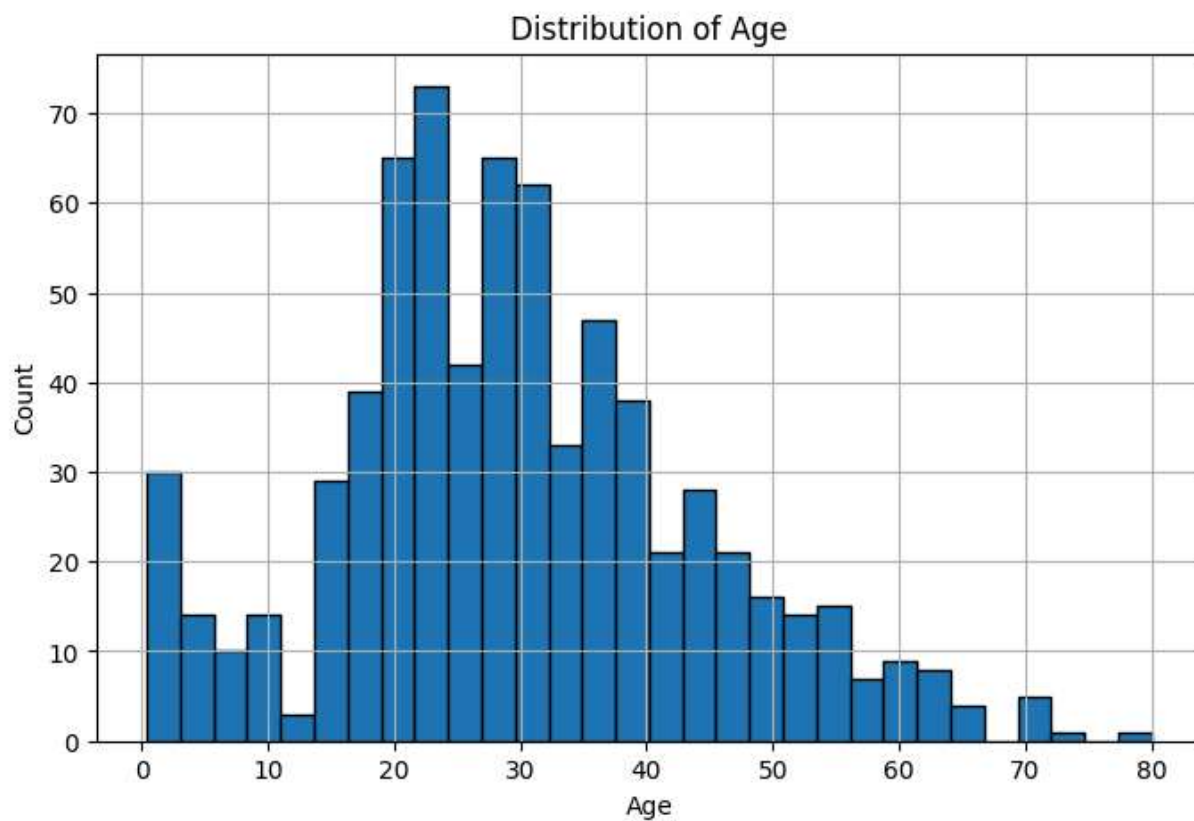
Q 77

Name: count, dtype: int64

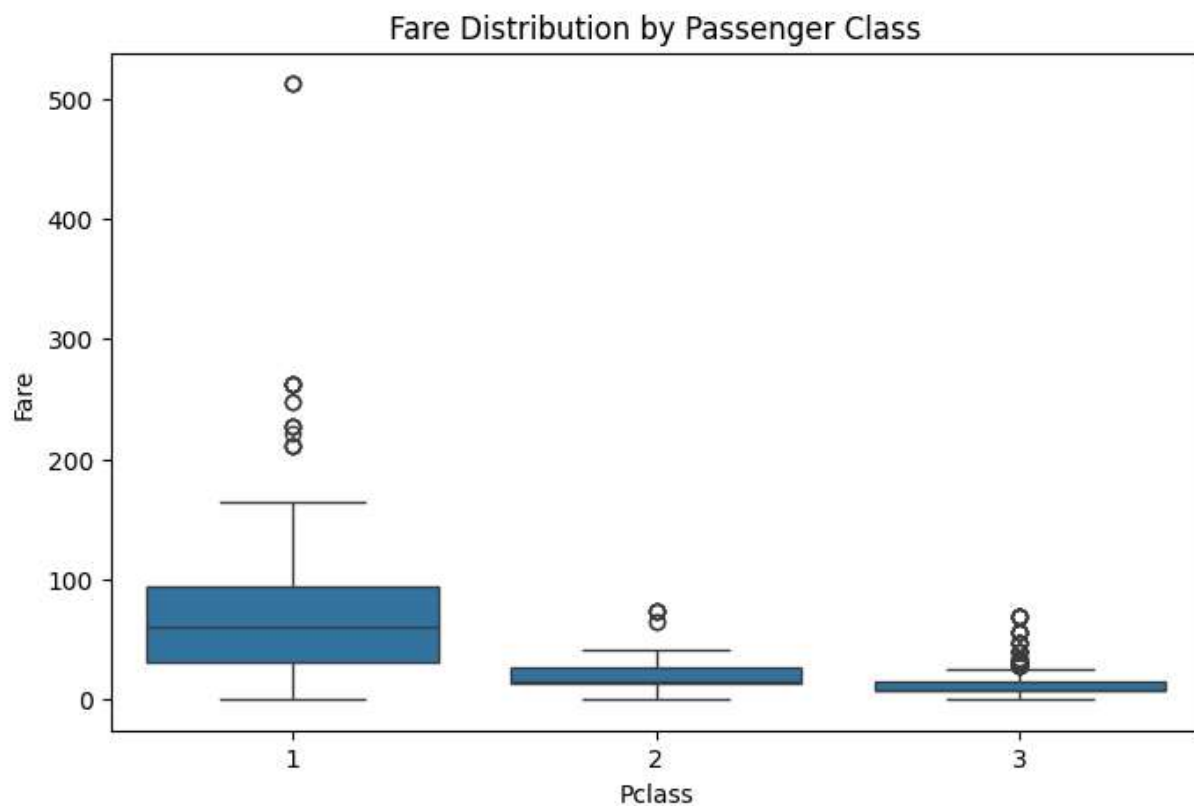
Univariate Analysis

Analyze distributions of individual variables.

```
In [21]: # Histogram of Age
plt.figure(figsize=(8,5))
titanic['Age'].hist(bins=30, edgecolor='black')
plt.title('Distribution of Age')
plt.xlabel('Age')
plt.ylabel('Count')
plt.show()
```



```
In [22]: # Boxplot of Fare by Passenger Class
plt.figure(figsize=(8,5))
sns.boxplot(x='Pclass', y='Fare', data=titanic)
plt.title('Fare Distribution by Passenger Class')
plt.show()
```



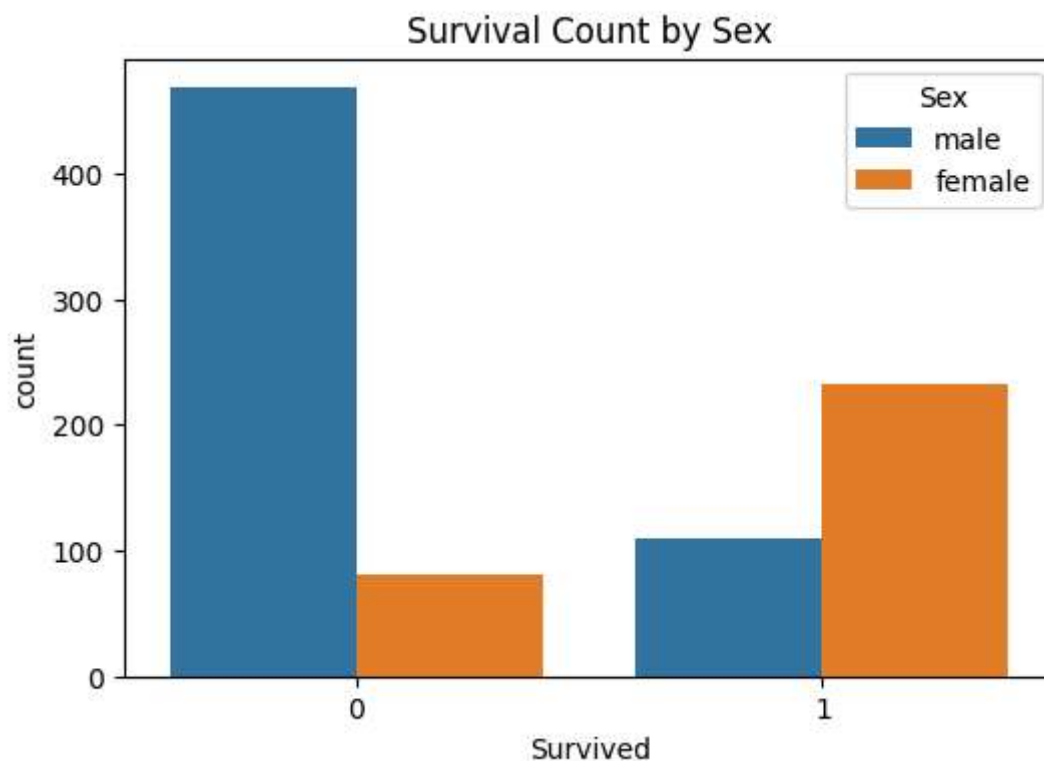
Observations:

- Most passengers are between ages 20 and 40.
- Fare prices vary significantly by passenger class, with some outliers in fare.

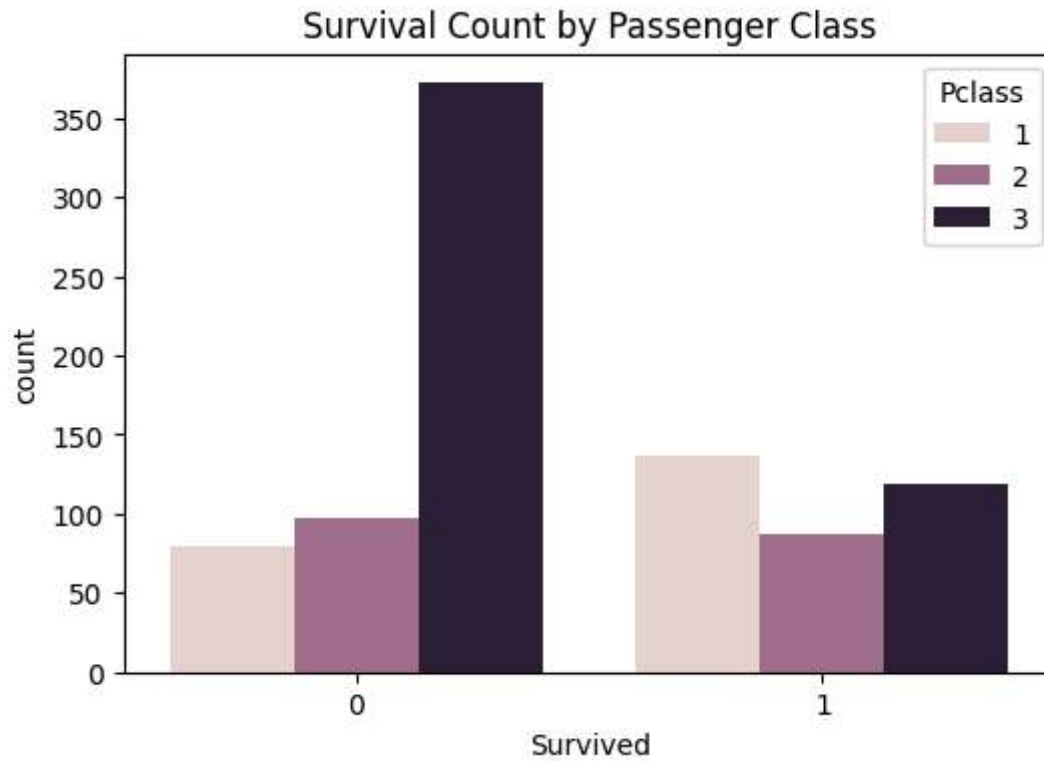
Bivariate Analysis

Explore relationships between two variables.

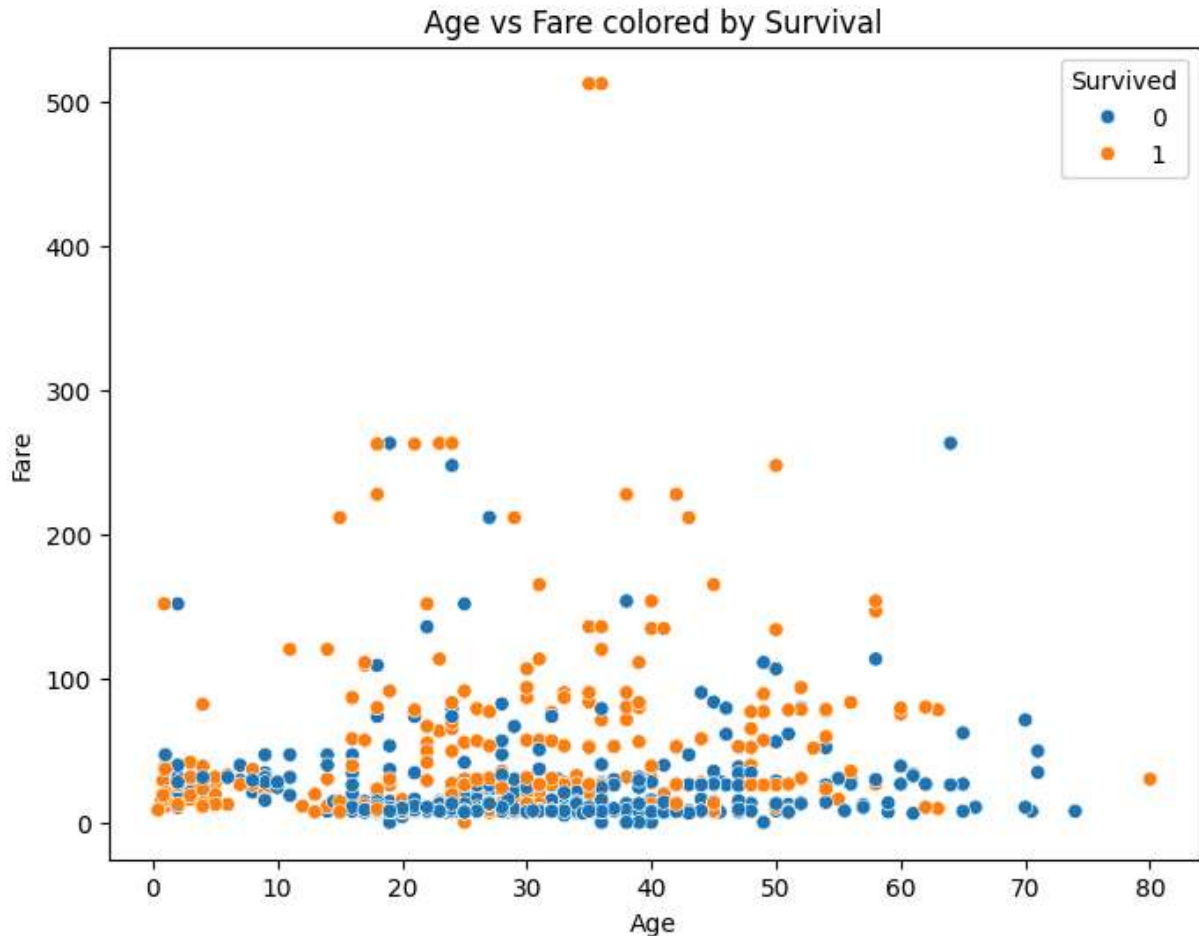
```
In [23]: # Survival count by Sex
plt.figure(figsize=(6,4))
sns.countplot(x='Survived', hue='Sex', data=titanic)
plt.title('Survival Count by Sex')
plt.show()
```



```
In [25]: # Survival count by Class
plt.figure(figsize=(6,4))
sns.countplot(x='Survived', hue='Pclass', data=titanic)
plt.title('Survival Count by Passenger Class')
plt.show()
```



```
In [27]: # Scatter plot Age vs Fare, colored by Survival
plt.figure(figsize=(8,6))
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=titanic)
plt.title('Age vs Fare colored by Survival')
plt.show()
```



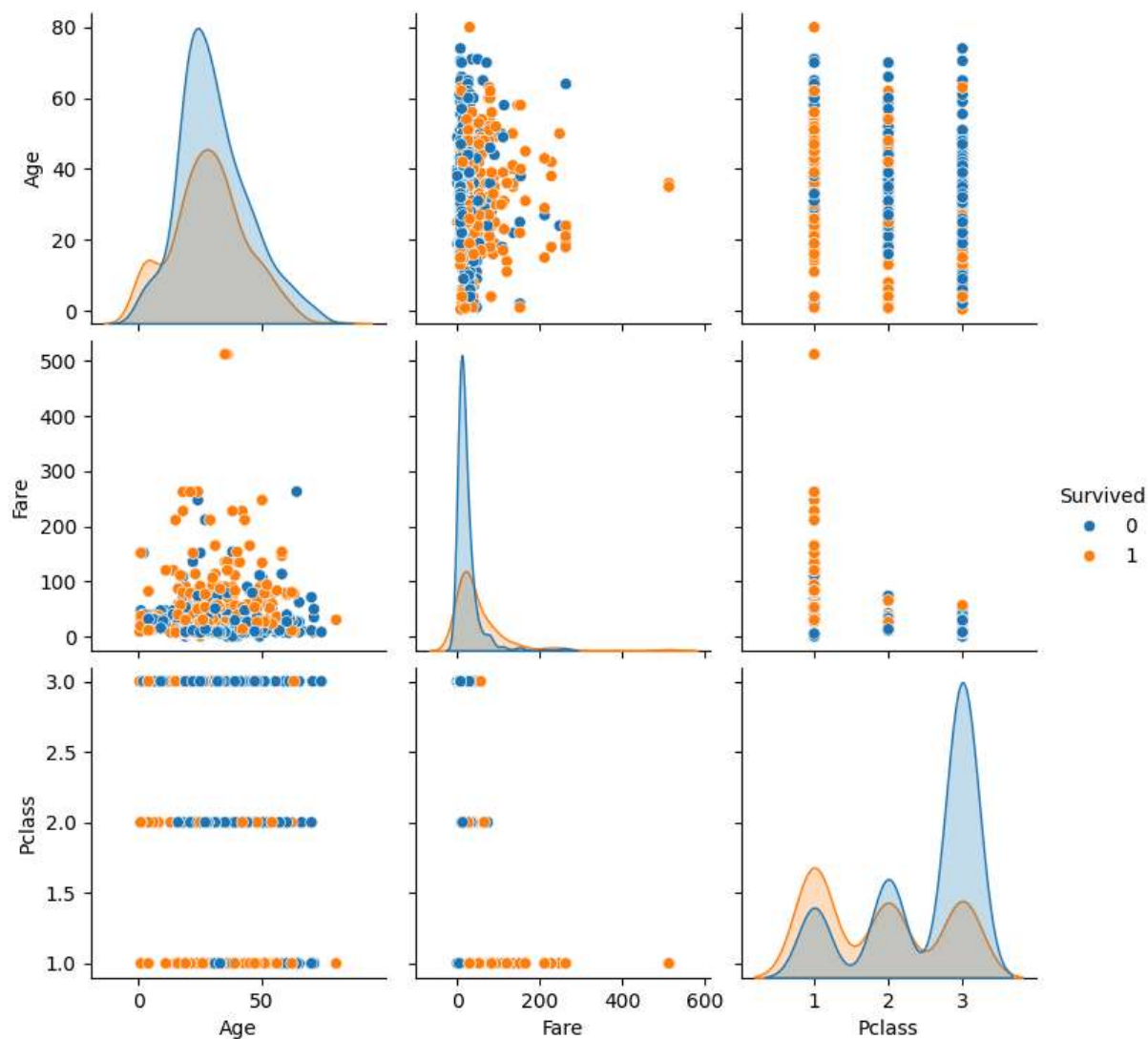
Observations:

- Females had higher survival rates compared to males.
- Passengers in first class had a higher survival rate.
- There is no strong visible relationship between age and fare concerning survival.

Multivariate Analysis

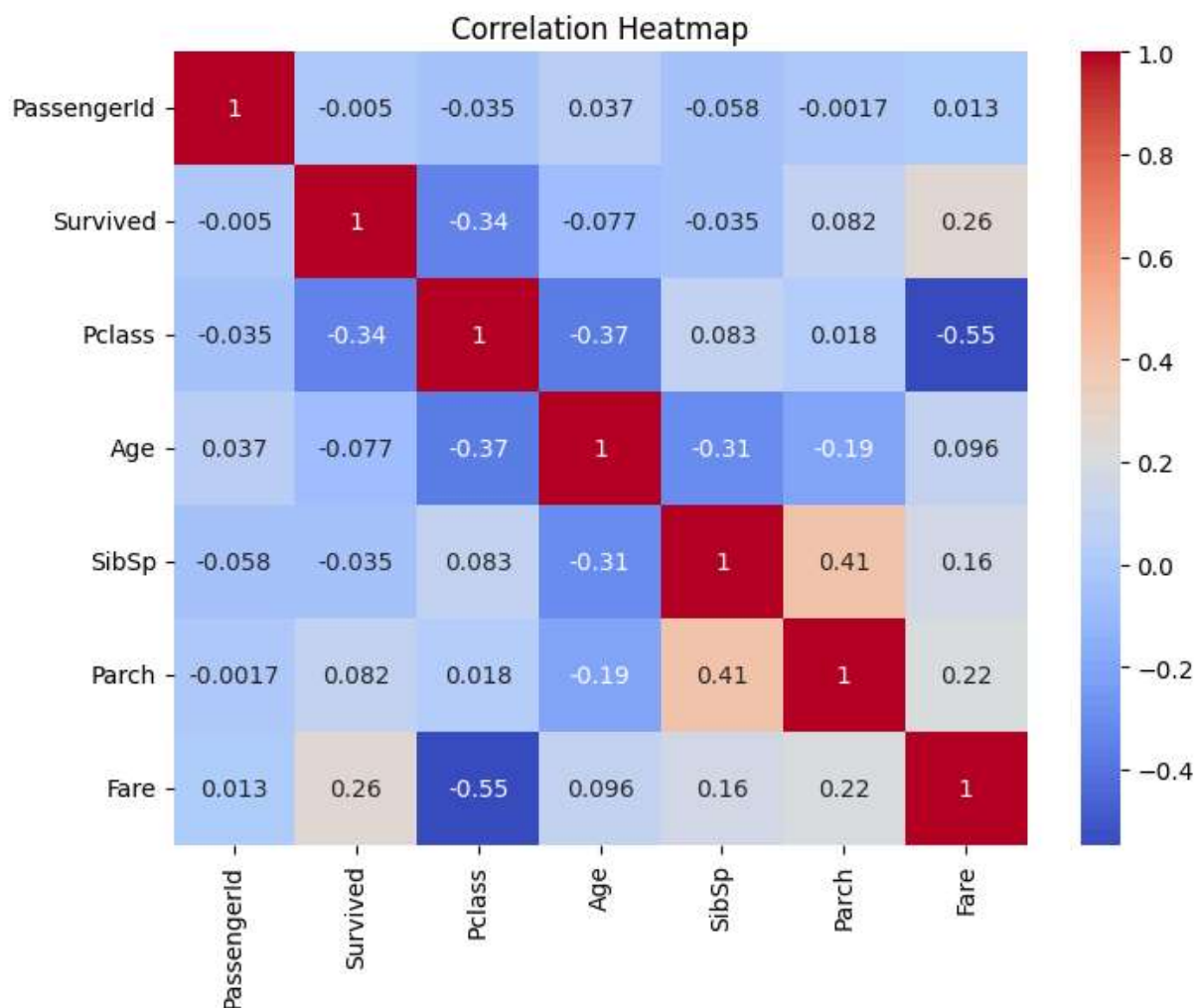
Check interactions among multiple variables.

```
In [28]: # Pairplot of selected numeric columns with survival hue
sns.pairplot(titanic[['Age', 'Fare', 'Pclass', 'Survived']].dropna(), hue='Survived')
plt.show()
```



```
In [31]: # Correlation Heatmap using only numeric columns
numeric_cols = titanic.select_dtypes(include=['float64', 'int64'])
corr = numeric_cols.corr()

plt.figure(figsize=(8,6))
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```



Observations:

- Fare negatively correlates with passenger class (higher class = higher fare).
- Survival weakly correlates with class and fare.
- Age does not show a strong correlation with survival.

Missing Data Analysis

Check how much data is missing and decide if imputation or removal is needed.

```
In [30]: # Count missing values per column
titanic.isnull().sum()
```

```
Out[30]: PassengerId      0
         Survived        0
         Pclass          0
         Name            0
         Sex             0
         Age            177
         SibSp           0
         Parch           0
         Ticket          0
         Fare            0
         Cabin          687
         Embarked        2
         dtype: int64
```

- Age and embarked ,Cabin columns contain missing data.
- Consider imputing or dropping rows for further analysis.

Summary of Findings

- **Survival** was strongly influenced by **sex** and **passenger class**; females and first-class passengers had higher survival rates.
- The **fare** paid varied by class, with first-class passengers paying much higher fares.
- Most passengers were aged between 20-40 years; age had less impact on survival.
- Missing data primarily in the **age** ,**Cabin** and **embarked** columns may affect some analyses.
- Correlation analysis shows a weak positive correlation between survival and fare/class.

This concludes the exploratory data analysis of the Titanic dataset.